

SMART CITY MANAGEMENT SYSTEM USING ESP32

Abstract

The Smart City Management System is an IoT-based project developed using the ESP32 microcontroller. The project integrates multiple smart city services into a single system, including smart traffic management, smart street lighting, smart waste management, water tank monitoring, air quality monitoring, and environmental monitoring. Various sensors collect real-time data, and the ESP32 processes this information to automate city operations. The OLED display provides live system status, making the system efficient, reliable, and suitable for modern smart city applications.

Introduction

Smart cities use IoT technology to improve public services and efficiently manage resources. This project demonstrates how an ESP32 can integrate multiple smart city functions into one system. It reduces manual intervention, enhances safety, conserves energy, and improves monitoring through automated decision-making.

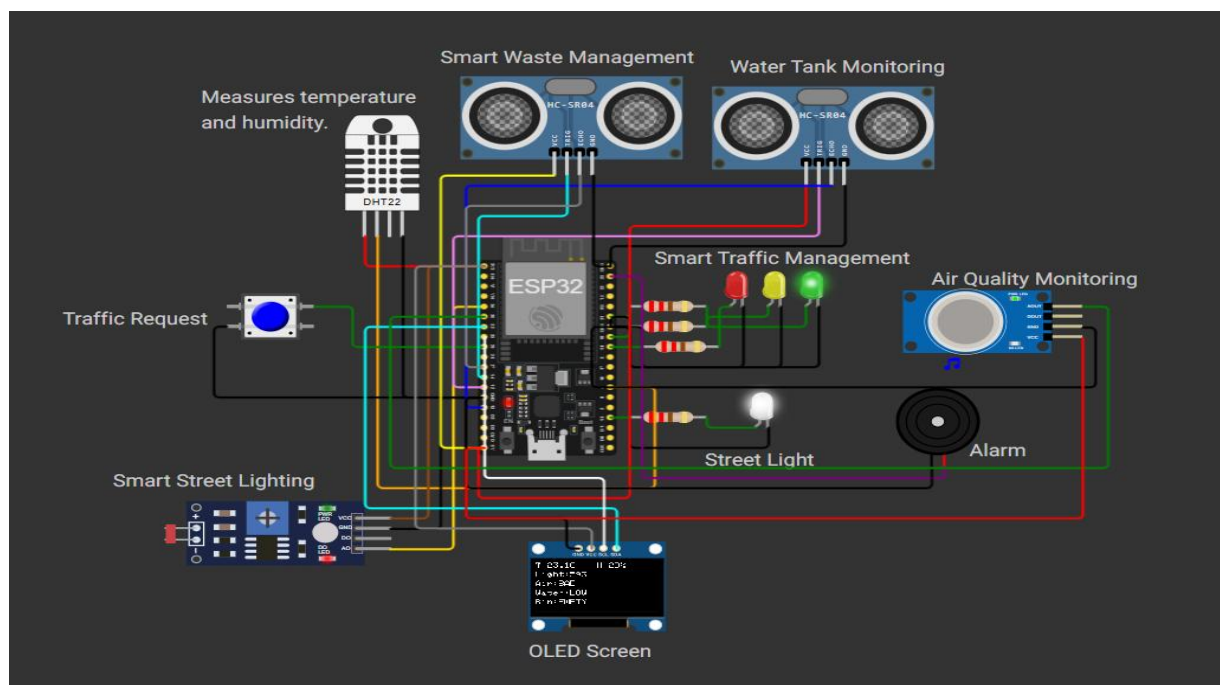
Objectives

- Develop an IoT-based Smart City Management System.
 - Automate traffic signal control.
 - Monitor waste bin levels.
 - Monitor water tank levels.
 - Monitor temperature and humidity.
 - Monitor air quality.
 - Control street lights automatically.
 - Display real-time information on an OLED display.
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Components Used

Component	Purpose
ESP32	Main controller
DHT22	Temperature and humidity monitoring
HC-SR04 (Waste)	Waste bin level detection
HC-SR04 (Water Tank)	Water level monitoring
MQ Gas Sensor	Air quality monitoring
LDR Module	Automatic street lighting
OLED Display	Display sensor data
Red, Yellow, Green LEDs	Traffic signal
Push Button	Traffic request input
Buzzer	Alarm for abnormal conditions
Jumper Wires	Connections

Circuit Diagram



Working Principle

The ESP32 continuously reads data from all connected sensors and controls different smart city modules.

Environmental Monitoring

The DHT22 sensor measures temperature and humidity. These values are displayed on the OLED screen for environmental monitoring.

Smart Waste Management

An HC-SR04 ultrasonic sensor measures the distance to the waste level inside the dustbin. When the measured distance indicates that the bin is nearly full, the OLED displays a "Bin Full" message.

Water Tank Monitoring

A second HC-SR04 ultrasonic sensor measures the water level in the storage tank. If the tank becomes full, the system activates the buzzer and updates the OLED display.

Smart Traffic Management

The traffic signal normally follows the sequence:

- Green LED
- Yellow LED
- Red LED

When the traffic request push button is pressed, the signal changes according to the programmed timing to allow vehicle movement.

Smart Street Lighting

The LDR module detects ambient light. During low-light conditions, the street light LED turns ON automatically. During daylight, it remains OFF.

Air Quality Monitoring

The MQ sensor continuously monitors air quality. If harmful gases exceed the threshold value, the buzzer is activated and an alert is displayed on the OLED.

OLED Display

The OLED screen continuously updates:

- Temperature
 - Humidity
 - Air Quality Status
 - Water Tank Status
 - Waste Bin Status
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Algorithm

1. Start ESP32.
 2. Initialize all sensors and OLED display.
 3. Read temperature and humidity.
 4. Read waste bin distance.
 5. Read water tank distance.
 6. Read air quality.
 7. Read LDR value.
 8. Control traffic lights.
 9. Control street light.
 10. Activate buzzer if required.
 11. Update OLED display.
 12. Repeat continuously.
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Applications

- Smart Cities
- Municipal Corporations
- Smart Buildings
- Industrial Monitoring

- Environmental Monitoring
 - Traffic Management Systems
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Advantages

- Reduces manual monitoring.
 - Saves electricity using automatic street lighting.
 - Improves traffic management.
 - Monitors air quality in real time.
 - Prevents water overflow.
 - Supports smart waste collection.
 - Low-cost IoT solution.
 - Easy to expand with cloud connectivity.
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Future Scope

- Mobile application integration.
 - Cloud-based monitoring.
 - AI-based traffic optimization.
 - Smart parking management.
 - CCTV integration.
 - GPS-based monitoring.
 - Voice notification system.
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Results

The Smart City Management System successfully simulated multiple city services using ESP32. The system monitored environmental conditions, traffic, water level, waste level, air quality, and street lighting while displaying all important information on the OLED screen. Automatic control

improved efficiency and demonstrated the practical application of IoT in smart city infrastructure.

Conclusion

The Smart City Management System using ESP32 successfully integrates several smart city modules into a single platform. The project demonstrates efficient automation through sensor-based monitoring and control. It improves resource management, enhances public safety, and supports sustainable urban development. This project provides a strong foundation for future IoT-based smart city applications.